

Online Internship on Open-Source Processor Architecture (IBM OpenPOWER) and SoC Design

Assignment No. 04: Sequential Logic Design

Instructions to Students

Please upload the complete compiled code, simulation screenshots, and documentation compiled as a single PDF file.

Assignment Questions

Q1. Design and Simulate a 4-bit Up/Down Counter using Verilog HDL.

- a) Behavioral modeling.
- b) Synchronous reset implementation.
- c) Asynchronous reset implementation.
- d) Verify up-counting operation.
- e) Verify down-counting operation.